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MID TERM EXAMINATION-I

B.TECH PROGRAMMES (UNDER THE AEGIS OF USAR)
SECOND Semester, March, 2026

Paper Code: BS112**Subject: Applied Mathematics-II****Time: 1½ Hrs.****Max. Marks: 30****Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.**

Q. No.	Question	Max. Marks	CO(s)
1. (a)	Define the following: (i) Analytic Function along with example (ii) Harmonic function	2.5	1
(b)	Determine whether $1/z$ is analytic or not.	2.5	1
(c)	Explain (i) Residue at point "a", (ii) Residue at Infinity.	2.5	2
(d)	Determine the poles of the following function and residue at each pole: $f(z) = z^2 / [(z - 1)^2(z + 2)]$ and hence evaluate integral $\int_C f(z) dz$, where $C: z = 3$.	2.5	2
2. (a)	Show that $u = x^4 - 6x^2y^2 + y^4$ and $v = y / (x^2 + y^2)$ are harmonic functions but are not harmonic conjugates.	5	1
(b)	Evaluate the following complex integral using Cauchy's integral formula: $\oint_C [(3z^2 + 7z + 1) / ((z^2 + 1)(z + 2))] dz$, where C is the circle $ z = 2$.	5	1
3. (a)	Evaluate $\int_C [e^z / \cos(\pi z)] dz$, where C is the unit circle $ z = 1$.	5	2
(b)	Evaluate $\int_0^\infty [\cos(mx) / (x^2 + 1)] dx$.	5	2
4. (a)	Show that the function $f(z)$ defined by: $f(z) = [x^3(1 + i) - y^3(1 - i)] / (x^2 + y^2)$, for $z \neq 0$, and $f(0) = 0$ is continuous and the Cauchy-Riemann equations are satisfied at the origin. Yet $f'(0)$ does not exist.	5	1
(b)	Find and plot the image of triangular region with vertices (0,0), (1,0), (0,1) under the transformation $w = (1 - i)z + 3$.	5	2

****** End of Question Paper ******